

Section B

| Question |     |       |  | Marking details                                                                                                                                                                                                                      | Marks available |     |     |       |       |      |
|----------|-----|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                      | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9        | (a) | (i)   |  | <b>C</b> (1)<br><br>contains two shells not three / less shielding of outer electron (1)<br><br>of the elements with two shells it contains greatest nuclear charge / most protons (1)                                               | 2               |     | 1   | 3     |       |      |
|          |     | (ii)  |  | <b>G</b> (1)<br><br>nucleus is most positively charged / has most protons (so greatest attraction to last electron) (1)                                                                                                              | 1               |     | 1   | 2     |       |      |
|          |     | (iii) |  | <b>D, E, G</b><br><br>award (2) for all three correct<br>award (1) for two correct / two correct and one incorrect answer<br>award (1) for all three correct and one incorrect answer<br>award (0) if more than one incorrect answer | 1               | 1   |     | 2     |       |      |
|          |     | (iv)  |  | <b>D, (E), G</b>                                                                                                                                                                                                                     | 1               |     |     | 1     |       |      |
|          | (b) | (i)   |  | (gaseous magnesium) atoms bombarded by electrons                                                                                                                                                                                     | 1               |     |     | 1     |       |      |
|          |     | (ii)  |  | deflected through a magnetic field                                                                                                                                                                                                   | 1               |     |     | 1     |       |      |